STACKS, SHEAVES ON ALGEBRAIC STACKS: PROPOSITION-COHERATOR

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2656-2676 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 2678-2748, the complete author proof (69 body lines).

Editorial note. Navigation stubs: 10 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 6 of queue_algebraic-stacks-sheaves.json (pb=2918, stmt 2656-2676, proof 2678-2748).

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'proposition-coherator' also exists in stacks-spaces-properties — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original proposition 001P: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2748–2748. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07D7: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2748–2748. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07D8: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2748–2748. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 079A: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2748–2748. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 079H: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2748–2748. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition lemma-quasi-coherent-algebraic-stack: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2601–2631. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 077P: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2748–2748. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 03DB: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2748–2748. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0780: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2748–2748. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 077V: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 2748–2748. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1. *Let $\mathcal{X}$ be an algebraic stack over S.*

(1) *The category $QCoh(\mathcal{O}_\mathcal{X})$ is a Grothendieck abelian category. Consequently, $QCoh(\mathcal{O}_\mathcal{X})$ has enough injectives and all limits.*
(2) *The inclusion functor $QCoh(\mathcal{O}_\mathcal{X}) \to Mod(\mathcal{O}_\mathcal{X})$ has a right adjoint*[1]
$$Q : Mod(\mathcal{O}_\mathcal{X}) \longrightarrow QCoh(\mathcal{O}_\mathcal{X})$$
such that for every quasi-coherent sheaf $\mathcal{F}$ the adjunction mapping $Q(\mathcal{F}) \to \mathcal{F}$ is an isomorphism.

Proof. This proof is a repeat of the proof in the case of schemes, see Properties, Proposition 077P and the case of algebraic spaces, see Properties of Spaces, Proposition 077V. We advise the reader to read either of those proofs first.

Part (1) means $QCoh(\mathcal{O}_\mathcal{X})$ (a) has all colimits, (b) filtered colimits are exact, and (c) has a generator, see Injectives, Section 079A. By Lemma lemma-quasi-coherent-algebraic-stack colimits in $QCoh(\mathcal{O}_X)$ exist and agree with colimits in $Mod(\mathcal{O}_X)$. By Modules on Sites, Lemma 03DB filtered colimits are exact. Hence (a) and (b) hold.

Choose a presentation $\mathcal{X} = [U/R]$ so that (U, R, s, t, c) is a smooth groupoid in algebraic spaces and in particular s and t are flat morphisms of algebraic spaces. By Lemma lemma-quasi-coherent-algebraic-stack above we have $QCoh(\mathcal{O}_\mathcal{X}) = QCoh(U, R, s, t, c)$. By Groupoids in Spaces, Lemma 0780 there exists a set T and a family $(\mathcal{F}_t)_{t\in T}$ of quasi-coherent sheaves on $\mathcal{X}$ such that every quasi-coherent sheaf on $\mathcal{X}$ is the directed colimit of its subsheaves which are isomorphic to one of the $\mathcal{F}_t$. Thus $\bigoplus_t \mathcal{F}_t$ is a generator of $QCoh(\mathcal{O}_X)$ and we conclude that (c) holds. The assertions on limits and injectives hold in any Grothendieck abelian category, see Injectives, Theorem 079H and Lemma 07D8.

[1] This functor is sometimes called the *coherator*.

Proof of (2). To construct Q we use the following general procedure. Given an object $\mathcal{F}$ of $Mod(\mathcal{O}_\mathcal{X})$ we consider the functor

$$QCoh(\mathcal{O}_\mathcal{X})^{opp} \longrightarrow Sets, \quad \mathcal{G} \longmapsto \mathrm{Hom}_\mathcal{X}(\mathcal{G}, \mathcal{F})$$

This functor transforms colimits into limits, hence is representable, see Injectives, Lemma 07D7. Thus there exists a quasi-coherent sheaf $Q(\mathcal{F})$ and a functorial isomorphism $\mathrm{Hom}_\mathcal{X}(\mathcal{G}, \mathcal{F}) = \mathrm{Hom}_\mathcal{X}(\mathcal{G}, Q(\mathcal{F}))$ for $\mathcal{G}$ in $QCoh(\mathcal{O}_\mathcal{X})$. By the Yoneda lemma (Categories, Lemma 001P) the construction $\mathcal{F} \rightsquigarrow Q(\mathcal{F})$ is functorial in $\mathcal{F}$. By construction Q is a right adjoint to the inclusion functor. The fact that $Q(\mathcal{F}) \to \mathcal{F}$ is an isomorphism when $\mathcal{F}$ is quasi-coherent is a formal consequence of the fact that the inclusion functor $QCoh(\mathcal{O}_\mathcal{X}) \to Mod(\mathcal{O}_\mathcal{X})$ is fully faithful. $\square$